MIL-P-11570C
30 July 1984
SUPERSEDING
MIL-P-11570B
16 October 1974

MILITARY SPECIFICATION

PYROMETERS, INDICATING, THERMOCOUPLES

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers direct reading, indicating thermocouple pyrometers with a temperature range up to 2700° F.

1.2 Classification. Pyrometers shall be of the following types and classes, as specified (see 6.2):

Type I - Portable.
Type II - Wall-mounted.
Class 1 - Range up to 1600° F.
Class 2 - Range up to 2300° F.
Class 3 - Range up to 2700° F.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation, Methods of.
MIL-M-16034	- Meters, Electrical-Indicating (Switchboard and Portable Types).

STANDARDS

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

HANDBOOKS

MILITARY

MIL-HDBK-759	- Military Standardization Handbook, HFE Design for Army Materiel.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practices for Commercial Packaging.
E 220 - Calibration of Thermocouples by Comparison Techniques.
E 230 - Temperature Electromotive Force (EMF) Tables for Thermocouples.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

MC96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to the Instrument Society of America, 67 Alexandria Drive, P.O. Box 12277, Research Triangle Park, NC 27709.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pyrometer shall consist essentially of a millivoltmeter and a thermocouple.

3.2 First article. The contractor shall furnish one or more pyrometers for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods and choice of design detail will produce pyrometers that comply with this specification. Examination and test shall be as specified in section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the first article examination and tests, as specified (see 6.2).

3.3 Material. Material shall be as specified herein. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

3.4 Meter. The meter shall be a millivoltmeter and the movement of the meter shall be shielded against demagnetization. Unless otherwise specified (see 6.2), an automatic reference junction compensator shall be provided with means for converting to manual operation. The meter shall conform to MIL-M-16034, type MR51W075DCMV with accuracies as specified in 3.8 and 3.9, and shall be contained in a lightweight, shockproof housing. The meter scale shall conform to MIL-HDBK-759, 1.2.3.1 through 1.2.3.1.5.2. The housing shall protect the meter and the reference junction compensator from atmospheric conditions normally encountered in foundries. The meter shall be permanently and legibly marked as to the thermocouple for which the meter is calibrated.

3.5 Thermocouple. The thermocouple shall consist of 2 dissimilar metal wires. Means for identifying the wires shall be provided in accordance with ANSI MC96.1. The thermocouple shall be either protected or metal sheathed as specified (see 6.2). The metal sheathed thermocouple shall have an exposed or grounded end, as specified (see 6.2). Protected thermocouples shall have the type and number of protection tubes provided as specified (see 6.2). A shielded or unshielded connecting block shall be provided for connecting the thermocouple to the millivoltmeter or to connecting wires or cables. The thermocouple shall be adjustable for use in any position from the horizontal through any angle to not less than 90 degrees downward. The length of the thermocouple shall be as specified (see 6.2).

3.5.1 Connecting wires. Connecting wires shall be composed of materials that are compatible to the thermocouple material as defined in MC96.1 with limits of error as specified in ASTM E 230.

3.6 Identification marking. Each pyrometer shall be identified in accordance with MIL-STD-130.

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3.7 Finishing. Each pyrometer and its case shall be finished in accordance with the contractor's standard practice.

3.8 Type I - portable. Type I pyrometers shall be an integral unit and shall be provided with a comfortable handle. Type I meters shall have an accuracy of plus or minus 1 percent of reading based on an impressed voltage input to an indicated reading.

3.9 Type II - wall-mounted. Type II wall-mounted pyrometers shall consist of a thermocouple and a meter connected by an electric cable. The meter housing shall be provided with a means for mounting on a wall. Unless otherwise specified (see 6.2), the length of the connecting cable shall be 10 feet. Type II meters shall have an accuracy of plus or minus 0.50 percent of reading based on an impressed voltage input to an indicated reading.

3.10 Class 1.

3.10.1 Meter. The meter shall have a scale range of 32° F to 1600° F.

3.10.2 Thermocouple. The thermocouple shall be type E in accordance with ANSI MC96.1 and its accuracy shall conform to ASTM E 230.

3.11 Class 2.

3.11.1 Meter. The meter shall have a scale range of 32° F to 2300° F.

3.11.2 Thermocouples. The thermocouples shall be type K in accordance with ANSI MC96.1 and its accuracy shall conform to ASTM E 230.

3.12 Class 3.

3.12.1 Meter. The meter shall have a scale range of 32° F to 2700° F.

3.12.2 Thermocouple. The thermocouple shall be type R in accordance with ANSI MC96.1. Thermocouple accuracy shall conform to ASTM E 230.

3.13 Workmanship. The pyrometers shall be free from such defects as cracks, scratches, dents, or other defects which could impair operation or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of preparation for delivery (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article pyrometers shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pyrometers shall be tested as specified in 4.5.2.1. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105, inspection level III.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined for the defects specified in 4.5.1. Acceptable quality level (AQL) shall be 1.5 percent defective.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1. AQL shall be 4.0 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. Pyrometers shall be examined for the following defects. Presence of one or more defects shall be cause for rejection.

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101. Meter not shielded against demagnetization.
102. Meter housing not as specified.
103. Type I pyrometers not provided with a handle.
104. Type II pyrometers not provided with a means for wall mounting.
105. Workmanship not as specified.
106. Identification marking not as specified, incomplete, or illegible.

4.5.2 Tests. Samples selected in accordance with 4.4.1 shall be subjected to the following test. Failure of the test shall be cause for rejection.

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4.5.2.1 Thermocouple. The thermocouple shall be tested in accordance with applicable provisions of ASTM E 220 with accuracy as defined in 3.10.2, 3.11.2 and 3.12.2. Nonconformance with these accuracy requirements shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Preservation not as specified for level A.
108.	Boxes not closed and sealed by method specified for level A preservation.
109.	Packing not as specified for level A or B.
110.	Preservation and packing not in accordance with the referenced document as specified for commercial.
111.	Marking missing, illegible, incorrect, or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Each complete pyrometer with the instructions shall be preserved in accordance with MIL-P-116, method IIb. The inner container shall be the contractor's standard carton or a close-fitting box conforming to PPP-B-636, type CF, class domestic, variety SW, grade 125. The outer container shall conform to PPP-B-636, W6c, or W5c, style optional. The box shall be closed and sealed as specified for method V in the appendix of the box specification.

5.1.2 Commercial. Each complete pyrometer with the instructions shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial, as specified (see 6.2).

5.2.1 Level A. Pyrometers of like description, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional. Box closure and strapping shall be in accordance with the appendix to the box specification.

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5.2.2 Level B. Pyrometers of like description, preserved as specified in 5.1, shall be packed as specified in 5.2.1, except that boxes shall be domestic type. As an alternate, boxes conforming to PPP-B-636, V3c, V11c, V13c, or V15c may be used when the quantity to be packed does not exceed the gross weight or size limitation of the box. Box closure and strapping shall be in accordance with appendix to the applicable box specification.

5.2.3 Commercial. Pyrometers, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition, weight and cube data shall be marked on the shipping container.

6. NOTES

6.1 Intended use. The pyrometers are intended for use in determining the temperature of molten metals. Type I pyrometer is intended for use where portability is required and type II is intended for use in a permanent location. Class 1 pyrometers are intended for use with tin, zinc alloys, lead alloys, and magnesium alloys. Class 2 pyrometers are for use with brass, bronze, and aluminum alloys. Class 3 pyrometers are for use with nickel-base alloys and steel alloys. The protected thermocouple is intended for use where the molten metal contains slag, dross, or other contaminating matter on the surface. The open-end thermocouple is for use with metals which are excellent conductors of electricity in the molten state. The closed-end thermocouple is for use with metals which are either good or poor conductors of electricity in the molten state.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of pyrometer required (see 1.2).
c. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
d. The time frame required for submission of first article and number of models required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When a reference junction compensator shall not be provided (see 3.4).
g. Whether the thermocouple wires are metal sheathed or protected (see 3.5).

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h. Whether the metal sheathed thermocouple shall have an exposed or grounded end (see 3.5).
i. Type and number of protection tubes required for the protected thermocouples (see 3.5).
j. Length of the thermocouple required (see 3.5).
k. Length of cable required if other than as specified in 3.9.
l. Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

6.3 First article. Any changes or deviations of production pyrometers from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pyrometers conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manual to be furnished with each pyrometer.

6.5 Classification change. Type I - Portable and Type II - Wall-mounted, were added.

6.6 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-11570C	Pyrometers, Indicating, Thermocouples

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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